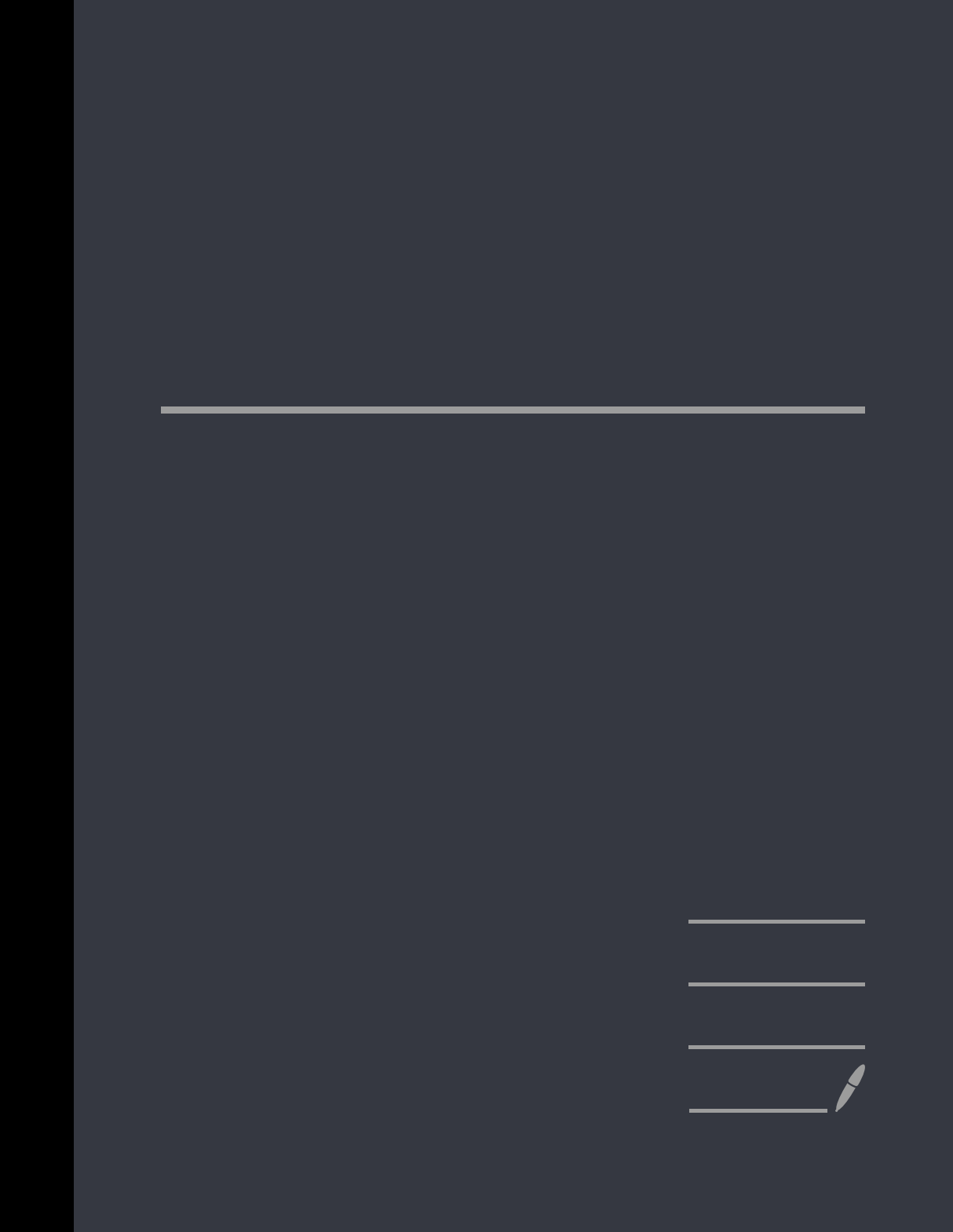


Eigenvektor der Matrix

$$A = \begin{pmatrix} 0 & 2 & -1 \\ 2 & -1 & 1 \\ 2 & -1 & 3 \end{pmatrix}$$

1. Berechnung charakteristische Polynom

$$\chi_A(\lambda) = \det(A - \lambda E_3) = \begin{vmatrix} 0-\lambda & 2 & -1 \\ 2 & -1-\lambda & 1 \\ 2 & -1 & 3-\lambda \end{vmatrix} = -\lambda^3 + 2\lambda^2 + 4\lambda - 8 = 0$$

NR: $\begin{vmatrix} -\lambda & 2 & -1 \\ 2 & -1-\lambda & 1 \\ 2 & -1 & 3-\lambda \end{vmatrix} \begin{matrix} -\lambda & 2 \\ 2 & -1-\lambda \\ 2 & -1 \end{matrix}$

$= -\lambda(-1-\lambda)\cdot(3-\lambda) + 2\cdot 1\cdot 2 + (-1)\,2\,(-1) - 2(-1-\lambda)\cdot(-1) - (-1)$

$= 1\left[(-1) - (3-\lambda)\cdot 2\cdot 2\right] = (\lambda+\lambda^2)(3-\lambda) + 4 + 2 - (2+2\lambda) - \lambda - (12 - 4\lambda)$

$= 3\lambda + 3\lambda^2 - \lambda^2 - \lambda^3 + 6 - 2 - 2\lambda - \lambda - 12 + 4\lambda$

$= -\lambda^3 + 2\lambda^2 + 4\lambda - 8$

Bestimmung der Eigenwerte als Nullstelle von $\chi_A(\lambda)$

$\chi_A(\lambda) = -\lambda^3 + 2\lambda^2 + 4\lambda - 8 = 0$

Nullstelle raten $\lambda = 2$ $\quad -8 + 8 + 8 - 8 = 0$

Polynomdivision: $(-\lambda^3 + 2\lambda^2 + 4\lambda - 8) : (\lambda - 2) = -\lambda^2 + 4$

$$\begin{array}{r} -\lambda^3 + 2\lambda^2 \\ \hline 4\lambda - 8 \\ -(4\lambda - 8) \\ \hline 0 \end{array}$$

$-\lambda^2+4=0 \quad |+\lambda^2$

$\lambda^2=4$

$\lambda_{1,2}=\pm 2$

Eigenwerte $\lambda_{1,2}=2 \quad \lambda_3=-2$

3. Bestimmung der Eigenvektoren durch Lösen des Gleichungssystem $(A-\lambda_i E_3)=0$

$\lambda_1=2 \quad (A-2E_3)|x = \begin{pmatrix} 0-2 & 2 & -1 \\ 2 & -1-2 & 1 \\ 2 & -1 & 3-2 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$

Löse das LGS

$$\left(\begin{array}{ccc|c} -2 & 2 & -1 & 0 \\ 2 & -3 & 1 & 0 \\ 2 & -1 & 1 & 0 \end{array}\right) \begin{array}{l} \\ |z_2+z_1 \\ |z_3+z_1 \end{array} \rightsquigarrow \left(\begin{array}{ccc|c} -2 & 2 & -1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{array}\right) |z_3+z_2 \rightsquigarrow \left(\begin{array}{ccc|c} -2 & 2 & -1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

Eigenvektor: $x = \begin{pmatrix} -\frac{\alpha}{2} \\ 0 \\ \alpha \end{pmatrix} = \alpha \begin{pmatrix} -1/2 \\ 0 \\ 1 \end{pmatrix} \quad \alpha \in \mathbb{R} \setminus \{0\}$

z_3: frei wählbar! $x_3 = \alpha \in \mathbb{R}$

z_2: $-x_2 = 0 \rightsquigarrow x_2 = 0$

z_1: $-2x_1 + 2x_2 - x_3 = 0$

$-2x_1 + 2\cdot 0 - \alpha = 0$

$-2x_1 + 0 - \alpha = 0 \quad |+\alpha$

$-2x_1 = \alpha \quad |:-2$

$x_1 = -\frac{\alpha}{2}$

Eigenraum: $E_A(2) = \{x \in \mathbb{R}^3 \mid Ax = 2x\} = \left\{ \begin{pmatrix} -\frac{\alpha}{2} \\ 0 \\ \alpha \end{pmatrix} \middle| \alpha \in \mathbb{R} \right\}$

$\lambda_3 = -2 \quad (A + 2E_3)\, x = 0$

$$\begin{pmatrix} 0+2 & 2 & -1 \\ 2 & -1+2 & 1 \\ 2 & -1 & 3+2 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Löse das LGS

$$\left(\begin{array}{ccc|c} 2 & 2 & -1 & 0 \\ 2 & 1 & 1 & 0 \\ 2 & -1 & 5 & 0 \end{array}\right) \begin{array}{l} \\ \\ \mid z_3 - z_1 \end{array} \rightsquigarrow \left(\begin{array}{ccc|c} 2 & 2 & -1 & 0 \\ 0 & -1 & 2 & 0 \\ 0 & -3 & 6 & 0 \end{array}\right) \begin{array}{l} \\ \\ \mid z_3 - 2z_2 \end{array} \left(\begin{array}{ccc|c} 2 & 2 & -1 & 0 \\ 0 & -1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

z_3: frei wählbar: $x_3 = a \in \mathbb{R}$

z_2: $-x_2 + 2x_3 = 0$

$x_2 + 2 \cdot a = 0 \quad | : 2a$

$x_2 = 2a$

z_1: $2x_1 + 2x_2 + x_3 = 0$

$2x_1 + 2 \cdot (-2a) + a = 0$

$2x_1 + 3a = 0 \quad | -3a$

$2x_1 = -3a \quad | : 2$

$x_1 = -\frac{3}{2}a$

Eigenvektor $= \begin{pmatrix} -\frac{3}{2}a \\ 2a \\ a \end{pmatrix} \quad a \in \mathbb{R} \setminus \{0\}$

Eigenraum: $E_A(-2) = \left\{ \begin{pmatrix} -\frac{3}{2}a \\ 2a \\ a \end{pmatrix} \middle| a \in \mathbb{R} \right\}$